



```
In [ ]: Reading Data from file
2. Schema of Dataframe
3. count of the number of Rows and Columns of Dataframe.
4. Show the player name and coach name
5. Show the records where Match ID must be equal to 1096 and calculate how many
6. order the data by match id
7. Show the records of position C and event is G40'.
8. Show the unique records with respect to match id.
9. Write a sql query to select player name and coach name from dataframe.
10. add a new column year which contains constant value "1930".
```

```
In [6]: from pyspark.sql import SparkSession
s = SparkSession.builder \
    .master("local[*]") \
    .appName("ColabSpark") \
    .getOrCreate()
```

```
In [54]: data=s.read.csv("mtcars.csv",header=True)
```

```
In [ ]: 2. Schema of Dataframe
```

```
In [10]: data.printSchema()
```

```
root
 |-- RoundID: string (nullable = true)
 |-- MatchID: string (nullable = true)
 |-- TeamInitials: string (nullable = true)
 |-- CoachName: string (nullable = true)
 |-- Line-up: string (nullable = true)
 |-- ShirtNumber: string (nullable = true)
 |-- PlayerName: string (nullable = true)
 |-- Position: string (nullable = true)
 |-- Event: string (nullable = true)
```

```
In [25]: print((data.count(), len(data.columns)))

(37784, 9)
```

```
In [ ]: Show the player name and coach name
```

```
In [28]: data.select("PlayerName", "CoachName").show()
```

PlayerName	CoachName
Alex THEPOT	CAUDRON Raoul (FRA)
Oscar BONFIGLIO	LUQUE Juan (MEX)
Marcel LANGILLER	CAUDRON Raoul (FRA)
Juan CARRENO	LUQUE Juan (MEX)
Ernest LIBERATI	CAUDRON Raoul (FRA)
Rafael GARZA	LUQUE Juan (MEX)
Andre MASCHINOT	CAUDRON Raoul (FRA)
Hilario LOPEZ	LUQUE Juan (MEX)
Etienne MATTTLER	CAUDRON Raoul (FRA)
Dionisio MEJIA	LUQUE Juan (MEX)
Marcel PINEL	CAUDRON Raoul (FRA)
Felipe ROSAS	LUQUE Juan (MEX)
Alex VILLAPLANE	CAUDRON Raoul (FRA)
Manuel ROSAS	LUQUE Juan (MEX)
Lucien LAURENT	CAUDRON Raoul (FRA)
Jose RUIZ	LUQUE Juan (MEX)
Marcel CAPELLE	CAUDRON Raoul (FRA)
Alfredo SANCHEZ	LUQUE Juan (MEX)
Augustin CHANTREL	CAUDRON Raoul (FRA)
Efrain AMEZCUA	LUQUE Juan (MEX)

only showing top 20 rows

In []: dShow the records where Match ID must be equal to 1096 and calculate how many

In [35]: a=data.filter(data.MatchID==1096)

In [39]: print(a.count())
print("-----")
a.show()

RoundID	MatchID	TeamInitials	CoachName	Line-up	ShirtNumber	Pl
ayerName	Position	Event				
x THEPOT	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Ale	
ONFIGLIO	201 1096	MEX LUQUE Juan (MEX)	S	0	Oscar B	
ANGILLER	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Marcel L	
CARRENO	201 1096	MEX LUQUE Juan (MEX)	S	0	Juan	
LIBERATI	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Ernest	
eI GARZA	201 1096	MEX LUQUE Juan (MEX)	S	0	Rafa	
ASCHINOT	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Andre M	
io LOPEZ	201 1096	MEX LUQUE Juan (MEX)	S	0	Hilar	
MATTLER	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Etienne	
io MEJIA	201 1096	MEX LUQUE Juan (MEX)	S	0	Dionis	
eI PINEL	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Marc	
pe ROSAS	201 1096	MEX LUQUE Juan (MEX)	S	0	Feli	
LLAPLANE	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Alex VI	
eI ROSAS	201 1096	MEX LUQUE Juan (MEX)	S	0	Manu	
LAURENT	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Lucien	
ose RUIZ	201 1096	MEX LUQUE Juan (MEX)	S	0	J	
CAPELLE	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Marcel	
SANCHEZ	201 1096	MEX LUQUE Juan (MEX)	S	0	Alfredo	
CHANTREL	201 1096	FRA CAUDRON Raoul (FRA)	S	0	Augustin	
AMEZCUA	201 1096	MEX LUQUE Juan (MEX)	S	0	Efrain	

only showing top 20 rows

In []: order the data by match id

```
In [40]: data.orderBy(data.MatchID).show()
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|RoundID|MatchID|TeamInitials|CoachName|Line-up|ShirtNumber|
|PlayerName|Position|Event|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 21| Z
denek HRUSKA| GK| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 18|
Petr JANECKA| NULL| 068'|
| 293| 1012| KUW| PARREIRA Carlos A...| S| 1| Ahmed
AL TARABULSI| GK| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 3|
Jan FIALA| NULL| NULL|
| 293| 1012| KUW| PARREIRA Carlos A...| S| 2|
Naim FAJAH| NULL| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 4| Ladis
lav JURKEMIK| NULL| NULL|
| 293| 1012| KUW| PARREIRA Carlos A...| S| 3| Mah
boud MUBARAK| NULL| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 5|
Jozef BARMOS| NULL| NULL|
| 293| 1012| KUW| PARREIRA Carlos A...| S| 5| Wali
d AL MUBARAK| NULL| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 8| Ant
onin PANENKA| NULL| P21'|
| 293| 1012| KUW| PARREIRA Carlos A...| S| 6| S
aad AL HUTTI| C| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 9| La
dislav VIZEK| NULL| NULL|
| 293| 1012| KUW| PARREIRA Carlos A...| S| 8| Abdulazi
z AL BALUSHI| NULL| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 10|
Tomas KRIZ| NULL| 061'|
| 293| 1012| KUW| PARREIRA Carlos A...| S| 9|
Jasem SULTAN| NULL| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 11| Z
denek NEHODA| C| NULL|
| 293| 1012| KUW| PARREIRA Carlos A...| S| 10| Abdulaz
iz AL ANBARI| NULL| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 13|
Jan BERGER| NULL| NULL|
| 293| 1012| KUW| PARREIRA Carlos A...| S| 14| Ab
dullah MAYUF| NULL| NULL|
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 15| J
ozef KUKUCKA| NULL| NULL|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
only showing top 20 rows
```

```
In [ ]: 7. Show the records of position C and event is G40'.
```

```
In [45]: data.filter((data.Position == 'c') & (data.Event == 'G40')).show()
```

```
+-----+-----+-----+-----+-----+-----+-----+
+-----+
|RoundID|MatchID|TeamInitials|CoachName|Line-up|ShirtNumber|PlayerName|Position|Event|
+-----+-----+-----+-----+-----+-----+-----+
+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+
```

```
In [46]: data.select('MatchId').distinct().show()
```

```
+-----+
|  MatchId|
+-----+
|    1090|
|    675|
| 43950027|
|300061497|
|    1372|
| 43950002|
|300061489|
|    919|
|    8763|
|300061465|
|300061494|
|300111113|
|    1953|
| 43950004|
|300061474|
|300186469|
|    475|
|    574|
|    3057|
|    8744|
+-----+
```

only showing top 20 rows

```
In [48]: data.dropDuplicates(['MatchId']).show()
```

```

+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|RoundID|MatchID|TeamInitials|CoachName|Line-up|ShirtNumber|
PlayerName|Position|Event|
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
| 293| 1012| TCH| VENGL0S Jozef (SVK)| S| 21| Zd
enek HRUSKA| GK| NULL|
| 323| 102| CMR| NEPOMNYASHCHI Val...| S| 16| T
homas NKON0| GK| NULL|
| 751| 103| ENG| ROBSON Bobby (ENG)| S| 1| Pe
ter SHILTON| GK| NULL|
| 293| 1044| YUG| MILJANIC Miljan (...| S| 1| Drag
an PANTELIC| GK| NULL|
| 293| 1051| SCO| STEIN Jock (SCO)| S| 1|
Alan ROUGH| GK| NULL|
| 293| 1054| URS| BESKOV Konstantin...| S| 1| R
inat DASAEV| GK| NULL|
| 293| 1055| POL| PIECHNICZEK Anton...| S| 1| Jozef
MLYNARCZYK| GK| NULL|
| 294| 1058| POL| PIECHNICZEK Anton...| S| 1| Jozef
MLYNARCZYK| GK| NULL|
| 293| 1071| URS| BESKOV Konstantin...| S| 1| R
inat DASAEV| GK| NULL|
| 322| 108| CMR| NEPOMNYASHCHI Val...| S| 16| T
homas NKON0| GK| Y68'|
| 201| 1084| ARG| OLAZAR Francisco ...| S| 0| A
ngel BOSSIO| GK| NULL|
| 201| 1085| ARG| OLAZAR Francisco ...| S| 0| A
ngel BOSSIO| GK| NULL|
| 201| 1086| ARG| OLAZAR Francisco ...| S| 0| A
ngel BOSSIO| GKC| NULL|
| 405| 1087| URU| SUPPICI Alberto (...| S| 0| Enrique
BALLESTRERO| GK| NULL|
| 202| 1088| ARG| OLAZAR Francisco ...| S| 0| J
uan BOTASSO| GK| NULL|
| 201| 1089| PAR| DURAND LAGUNA Jos...| S| 0| Pe
dro BENITEZ| GK| NULL|
| 201| 1090| USA| MILLAR Bob (USA)| S| 0| Ji
mmy DOUGLAS| GK| NULL|
| 201| 1091| BRA| DE CARVALHO Pinda...| S| 0|
VELLOSO| GK| NULL|
| 201| 1092| YUG| SIMONOVIC Bosko (...| S| 0| Mil
ovan JAKSIC| GK| NULL|
| 201| 1093| YUG| SIMONOVIC Bosko (...| S| 0| Mil
ovan JAKSIC| GK| NULL|
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
only showing top 20 rows

```

```
In [ ]: Write a sql query to select player name and coach name from dataframe.
```

```
In [49]: data.createOrReplaceTempView("EMP")
```

```
spark.sql("Select CoachName,PlayerName from Emp ").show()
```

```
+-----+-----+
|      CoachName|      PlayerName|
+-----+-----+
|CAUDRON Raoul (FRA)|      Alex THEPOT|
|  LUQUE Juan (MEX)|    Oscar BONFIGLIO|
|CAUDRON Raoul (FRA)|    Marcel LANGILLER|
|  LUQUE Juan (MEX)|      Juan CARRENO|
|CAUDRON Raoul (FRA)|    Ernest LIBERATI|
|  LUQUE Juan (MEX)|      Rafael GARZA|
|CAUDRON Raoul (FRA)|    Andre MASCHINOT|
|  LUQUE Juan (MEX)|      Hilario LOPEZ|
|CAUDRON Raoul (FRA)|    Etienne MATTLER|
|  LUQUE Juan (MEX)|      Dionisio MEJIA|
|CAUDRON Raoul (FRA)|      Marcel PINEL|
|  LUQUE Juan (MEX)|      Felipe ROSAS|
|CAUDRON Raoul (FRA)|    Alex VILLAPLANE|
|  LUQUE Juan (MEX)|      Manuel ROSAS|
|CAUDRON Raoul (FRA)|      Lucien LAURENT|
|  LUQUE Juan (MEX)|        Jose RUIZ|
|CAUDRON Raoul (FRA)|      Marcel CAPELLE|
|  LUQUE Juan (MEX)|    Alfredo SANCHEZ|
|CAUDRON Raoul (FRA)|    Augustin CHANTREL|
|  LUQUE Juan (MEX)|      Efrain AMEZCUA|
+-----+-----+
only showing top 20 rows
```

```
In [ ]: add a new column year which contains constant value "1930"
```

```
In [53]: from pyspark.sql.functions import lit
df=data.withColumn("year", lit(1930))

df.show()
```

RoundID	MatchID	TeamInitials	CoachName	Line-up	ShirtNumber	PL
ayerName	Position	Event	year			
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Ale	
x THEPOT	GK	NULL 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	Oscar B	
ONFIGLIO	GK	NULL 1930				
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Marcel L	
ANGILLER	NULL	G40' 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	Juan	
CARRENO	NULL	G70' 1930				
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Ernest	
LIBERATI	NULL	NULL 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	Rafa	
eI GARZA	C	NULL 1930				
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Andre M	
ASCHINOT	NULL G43'	G87' 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	Hilar	
io LOPEZ	NULL	NULL 1930				
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Etienne	
MATTLER	NULL	NULL 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	Dionis	
io MEJIA	NULL	NULL 1930				
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Marc	
eI PINEL	NULL	NULL 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	Feli	
pe ROSAS	NULL	NULL 1930				
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Alex VI	
LLAPLANE	C	NULL 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	Manu	
eI ROSAS	NULL	NULL 1930				
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Lucien	
LAURENT	NULL	G19' 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	J	
ose RUIZ	NULL	NULL 1930				
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Marcel	
CAPELLE	NULL	NULL 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	Alfredo	
SANCHEZ	NULL	NULL 1930				
201	1096	FRA CAUDRON Raoul (FRA)	S	0	Augustin	
CHANTREL	NULL	NULL 1930				
201	1096	MEX LUQUE Juan (MEX)	S	0	Efrain	
AMEZCUA	NULL	NULL 1930				

only showing top 20 rows

In [56]: `data.show()`

model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21	6	160	110	3.9	2.62	16.46	0	1	4	4
Mazda RX4 Wag	21	6	160	110	3.9	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.32	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.44	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.46	20.22	1	0	3	1
Duster 360	14.3	8	360	245	3.21	3.57	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.19	20	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.15	22.9	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.44	18.3	1	0	4	4
Merc 280C	17.8	6	167.6	123	3.92	3.44	18.9	1	0	4	4
Merc 450SE	16.4	8	275.8	180	3.07	4.07	17.4	0	0	3	3
Merc 450SL	17.3	8	275.8	180	3.07	3.73	17.6	0	0	3	3
Merc 450SLC	15.2	8	275.8	180	3.07	3.78	18	0	0	3	3
Cadillac Fleetwood	10.4	8	472	205	2.93	5.25	17.98	0	0	3	4
Lincoln Continental	10.4	8	460	215	3	5.424	17.82	0	0	3	4
Chrysler Imperial	14.7	8	440	230	3.23	5.345	17.42	0	0	3	4
Fiat 128	32.4	4	78.7	66	4.08	2.2	19.47	1	1	4	1
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.9	1	1	4	1

only showing top 20 rows

- In []:
1. Find the top five records of mpg attribute.
 2. Find out the records where the value of mpg is less than 18.
 3. Add a new column named as "wtTon" to dataframe with values of weight column multiplied by 0.45.
 4. Compute the average weight of cars by their cylinders.
 5. Write a sql query to find gears where the value of cylinder lies between 4 and 9.

In []: 1. Find the top five records of mpg attribute

In [59]: data.orderBy("mpg").show()

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
	Cadillac Fleetwood	10.4	8	472	205	2.93	5.25	17.98	0	0	3	4
	Lincoln Continental	10.4	8	460	215	3	5.424	17.82	0	0	3	4
	Camaro Z28	13.3	8	350	245	3.73	3.84	15.41	0	0	3	4
	Duster 360	14.3	8	360	245	3.21	3.57	15.84	0	0	3	4
	Chrysler Imperial	14.7	8	440	230	3.23	5.345	17.42	0	0	3	4
	Maserati Bora	15	8	301	335	3.54	3.57	14.6	0	1	5	8
	Merc 450SLC	15.2	8	275.8	180	3.07	3.78	18	0	0	3	3
	AMC Javelin	15.2	8	304	150	3.15	3.435	17.3	0	0	3	2
	Dodge Challenger	15.5	8	318	150	2.76	3.52	16.87	0	0	3	2
	Ford Pantera L	15.8	8	351	264	4.22	3.17	14.5	0	1	5	4
	Merc 450SE	16.4	8	275.8	180	3.07	4.07	17.4	0	0	3	3
	Merc 450SL	17.3	8	275.8	180	3.07	3.73	17.6	0	0	3	3
	Merc 280C	17.8	6	167.6	123	3.92	3.44	18.9	1	0	4	4
	Valiant	18.1	6	225	105	2.76	3.46	20.22	1	0	3	1
	Hornet Sportabout	18.7	8	360	175	3.15	3.44	17.02	0	0	3	2
	Merc 280	19.2	6	167.6	123	3.92	3.44	18.3	1	0	4	4
	Pontiac Firebird	19.2	8	400	175	3.08	3.845	17.05	0	0	3	2
	Ferrari Dino	19.7	6	145	175	3.62	2.77	15.5	0	1	5	6
	Mazda RX4	21	6	160	110	3.9	2.62	16.46	0	1	4	4
	Mazda RX4 Wag	21	6	160	110	3.9	2.875	17.02	0	1	4	4

only showing top 20 rows

In []: 2. Find out the records where the value of mpg is less than 18.

```
In [67]: data1=data.withColumn("Mpg",data.mpg.cast("double"))
data1.filter(data1.Mpg < 18).show()
```

	model	Mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
	Duster 360	14.3	8	360	245	3.21	3.57	15.84	0	0	3	4
	Merc 280C	17.8	6	167.6	123	3.92	3.44	18.9	1	0	4	4
	Merc 450SE	16.4	8	275.8	180	3.07	4.07	17.4	0	0	3	3
	Merc 450SL	17.3	8	275.8	180	3.07	3.73	17.6	0	0	3	3
	Merc 450SLC	15.2	8	275.8	180	3.07	3.78	18	0	0	3	3
	Cadillac Fleetwood	10.4	8	472	205	2.93	5.25	17.98	0	0	3	4
	Lincoln Continental	10.4	8	460	215	3	5.424	17.82	0	0	3	4
	Chrysler Imperial	14.7	8	440	230	3.23	5.345	17.42	0	0	3	4
	Dodge Challenger	15.5	8	318	150	2.76	3.52	16.87	0	0	3	2
	AMC Javelin	15.2	8	304	150	3.15	3.435	17.3	0	0	3	2
	Camaro Z28	13.3	8	350	245	3.73	3.84	15.41	0	0	3	4
	Ford Pantera L	15.8	8	351	264	4.22	3.17	14.5	0	1	5	4
	Maserati Bora	15.0	8	301	335	3.54	3.57	14.6	0	1	5	8

In [61]: data.printSchema()

```

root
|-- model: string (nullable = true)
|-- mpg: string (nullable = true)
|-- cyl: string (nullable = true)
|-- disp: string (nullable = true)
|-- hp: string (nullable = true)
|-- drat: string (nullable = true)
|-- wt: string (nullable = true)
|-- qsec: string (nullable = true)
|-- vs: string (nullable = true)
|-- am: string (nullable = true)
|-- gear: string (nullable = true)
|-- carb: string (nullable = true)

```

In []: Add a new column named **as** "wtTon" to dataframe **with** values of weight column multiplied by 0.45

```
In [ ]: data = data.withColumn("wtTon", data.wt * 0.45)
data.show()
```

In []: 4. Compute the average weight of cars by their cylinders.

```
In [70]: from pyspark.sql.functions import avg
data = data.withColumn("weight", data.wt.cast("double"))

data.groupBy("cyl").agg(avg("weight").alias("avg_weight")).show()
```

```

+---+-----+
|cyl|      avg_weight|
+---+-----+
|  8|3.9992142857142867|
|  6| 3.117142857142857|
|  4| 2.285727272727273|
+---+-----+

```

In []: 5. Write a sql query to find gears where the value of cylinder lies between 4 and 9.

```
In [78]: data.createGlobalTempView("ex1")
```

```
In [ ]: s.sql("""
SELECT gear
FROM ex1
WHERE cylinder IN ('4','9')
""").show()
```

In []:

In []: